

COAL Lab Project

Gear Installation.

Fair warning: installing these two libraries is where many students get stuck — not because it's hard, but because there are **multiple versions floating around** and the wrong one causes errors.

I'll give you the **clean, working method**. So everyone must use the **same**.

Step-by-step Installation (Arduino IDE)

Step 1: Install ESP32 Board (if not already)

1. Open Arduino IDE
2. Go to: **File → Preferences**
3. In Additional Board Manager URLs, add:
https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
4. Then: **Tools → Board → Boards Manager**
5. Search ESP32
6. Install esp32 by Espressif Systems

Note: Be patient, it will take some time.

Step 2: Install AsyncTCP

Method (Recommended – GitHub ZIP)

1. Go to GitHub: <https://github.com/me-no-dev/AsyncTCP.git>
2. Download ZIP of: **AsyncTCP**
3. In Arduino IDE:
Sketch → Include Library → Add .ZIP Library

4. Select downloaded ZIP

Note: Be patient, it will take some time.

Step 3: Install ESPAsyncWebServer

Same method:

1. Download ZIP of **ESPAsyncWebServer** from:
<https://github.com/me-no-dev/ESPAsyncWebServer.git>
2. Install via:
Sketch → Include Library → Add .ZIP Library

Note: Be patient, it will take some time.

IMPORTANT (Most Common Mistake)

These libraries are **NOT** officially maintained anymore
Some versions are **broken with newer ESP32 cores**

Test Installation

1. Upload this simple code:

```
#include <WiFi.h>
#include <AsyncTCP.h>
#include <ESPAsyncWebServer.h>
#include <FS.h>

AsyncWebServer server(80);
void setup() {
  Serial.begin(115200);
  WiFi.begin("test", "test");
  server.on("/", HTTP_GET, [](AsyncWebServerRequest *request){
    request->send(200, "text/plain", "Server working!");
  });
  server.begin();
}
void loop() {
}
```

2. Select **ESP32 Dev Module** and click verify:



If this compiles → you're good

Use Compatible Versions

If you face errors, do this:

Downgrade ESP32 board package:

Use version: 2.0.14 (very stable)

But in a group project every member should install the same version. And this setup is running properly in my device.

Common Errors + Fixes

“AsyncTCP.h not found”

➡ Library not installed properly

“multiple definition / compatibility errors”

➡ Version mismatch → fix by:

Downgrading ESP32 board

OR

Using updated library forks

“Compilation fails randomly”

➡ Restart Arduino IDE after installing libraries